

Multilayer Feed-Forward Neural Network

It is an interconnected Artificial Neural Network with multiple layers that has neurons with weights associated with them and they compute the result using activation functions. It is one of the types of Neural Networks in which the flow of the network is from input to output units and it does not have any loops, no feedback, and no signal moves in backward directions that is from output to hidden and input layer.

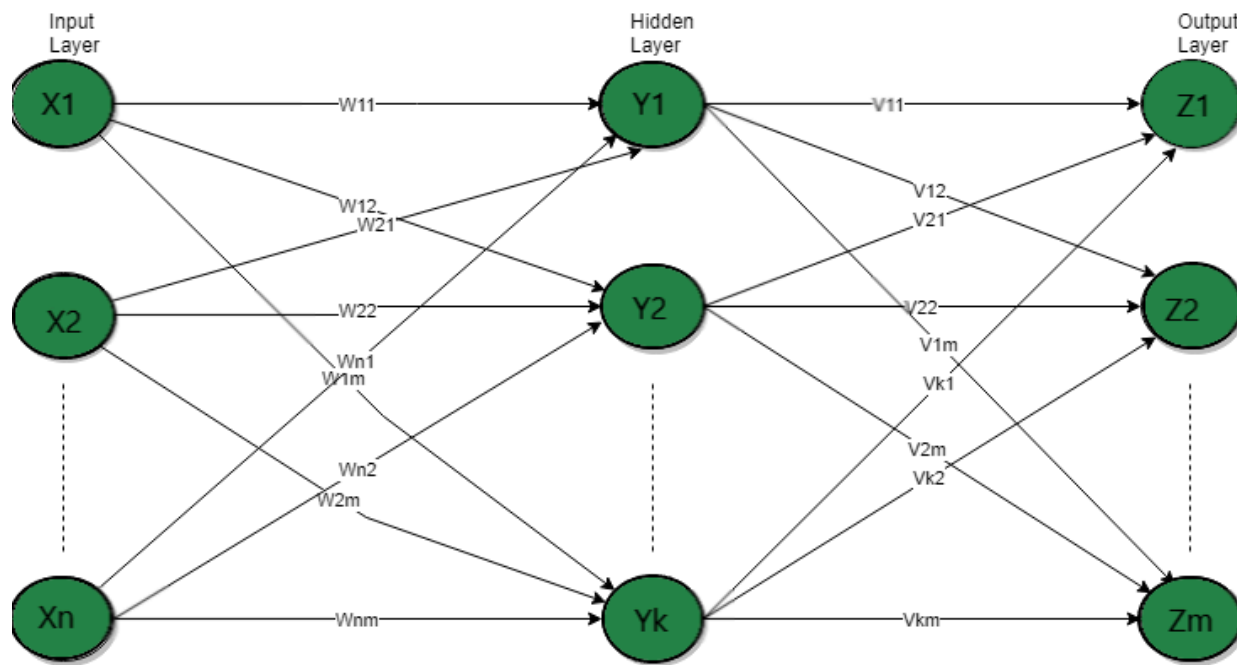
The ANN is a self-learning network that learns from sample data sets and signals, it is based on the function of the biological nervous system. The type of activation function depends on the desired output. It is a part of machine learning and AI, which are the fastest-growing fields, and lots of research is going on to make it more effective.

The Architecture of the Multilayer Feed-Forward Neural Network:

This Neural Network or Artificial Neural Network has multiple hidden layers that make it a multilayer neural Network and it is feed-forward because it is a network that follows a top-down approach to train the network. In this network there are the following layers:

1. **Input Layer:** It is the starting layer of the network that has a weight associated with the signals.
2. **Hidden Layer:** This layer lies after the input layer and contains multiple neurons that perform all computations and pass the result to the output unit.
3. **Output Layer:** It is a layer that contains output units or neurons and receives processed data from the hidden layer, if there are further hidden layers connected to it then it passes the weighted unit to the connected hidden layer for further processing to get the desired result.

The input and hidden layers use sigmoid and linear activation functions whereas the output layer uses a Heaviside step activation function at nodes because it is a two-step activation function that helps in predicting results as per requirements. All units also known as neurons have weights and calculation at the hidden layer is the summation of the dot product of all weights and their signals and finally the sigmoid function of the calculated sum. Multiple hidden and output layer increases the accuracy of the output.



Application of Multilayer Feed-Forward Neural Network:

1. Medical field
2. Speech regeneration
3. Data processing and compression
4. Image processing

Limitations:

This ANN is a basic form of Neural Network that has no cycles and computes only in the forward direction. It has some limitations like sometimes information about the neighborhood is lost and in that case, it becomes difficult to process further all steps needed to be performed again and it does not support back propagation so the network cannot learn or correct the fault of the previous stage.